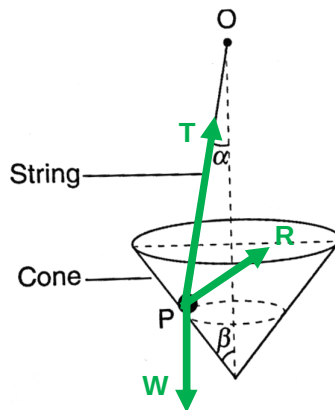


- 10 A pendulum bob at the end of the string which is fixed at O moves in a horizontal circle on the inside of a cone as shown.



If the tension in the string is T and the normal contact force by the cone is R , which of the following gives the centripetal force on the bob?

- A** $T \sin \alpha + R \cos \beta$
- B** $T \sin \alpha - R \cos \beta$
- C** $T \cos \alpha + R \sin \beta$
- D** $T \cos \alpha - R \sin \beta$

- 11 Two satellites X and Y are orbiting at heights of $5R$ and $3R$ above the surface of the Earth